

6G-Enabled Consumer Electronics Device Intrusion Detection With Federated Meta-Learning and Digital Twins in a Meta-Verse Environment

Suli He, Chengwen Du, and M. Shamim Hossain[✉], *Senior Member, IEEE*

Abstract—The widespread adoption of consumer electronics devices coupled with the emergence of 6G technology has led to the establishment of an extensive network of interconnected devices, forming the underlying infrastructure of the Internet of Things (IoT). Nevertheless, this interconnectivity introduces a myriad of security concerns, given that these devices become susceptible to malicious activities and unauthorized breaches. Moreover, conventional intrusion detection systems encounter difficulties in managing imbalanced data scenarios, wherein the count of normal instances vastly exceeds that of intrusion instances. To address this issue, we propose a novel framework for 6G-enabled consumer electronics device intrusion detection, leveraging the power of federated meta-learning and digital twins within a Meta-Verse environment. By leveraging the distributed intelligence of meta-learning across a network of devices, our framework enables efficient and accurate detection of intrusions while mitigating the impact of imbalanced data. Furthermore, by utilizing digital twins within a Meta-Verse environment, we create a scalable and controlled setting for experimentation, enabling the development and evaluation of intrusion detection algorithms in a realistic yet controlled manner. Our experimental results demonstrate the effectiveness of the proposed framework in detecting intrusions on 6G-enabled consumer electronics devices. The federated meta-learning approach achieves superior performance compared to traditional intrusion detection methods, especially in imbalanced data scenarios.

Index Terms—Federated meta learning, intrusion detection, metaverse, meta learning, digital twin, consumer electronics.

I. INTRODUCTION

THE BURGEONING Computer, Communication, and Consumer (3C) electronics industry is being propelled by the escalating demand for consumer electronic devices, including smartphones, tablets, and wearable consumer products [1], [2]. The emergence of the 6G network has transformed the connectivity landscape for 3C electronic devices, affording faster speeds, diminished latency, and augmented capacity [3], [4]. In conjunction with the pervasive integration of Internet

of Things (IoT)-enabled Consumer Electronics (CE) devices across diverse sectors like healthcare, transportation, and smart cities, the 6G network has grown in significance, ensuring dependable and secure connectivity [5]. Nevertheless, the adoption of IoT-enabled CE devices introduces novel security challenges, exposing the network to potential malevolent attacks. Of particular note, Digital Twins (DTs) [6], [7], acting as virtual replicas of physical entities, have gained paramount importance within the Metaverse by enabling cost-effective simulation and optimization of physical systems [8]. Yet, the sensitive data emanating from DTs and the vulnerability of these virtual entities to attacks have engendered security apprehensions [9].

Digital twin-enabled 6G Consumer Electronics Devices (CEDs) hold the potential to revolutionize our interaction with the physical world, enabling real-time monitoring, analysis, and control of metaverse systems [10]. However, the escalating proliferation of CEDs and digital twins introduces a concomitant escalation in data privacy and security risks. The substantial volume of data amassed by CEDs and digital twins can encompass sensitive information, comprising personal and corporate data, thereby rendering them susceptible to cyberattacks and unauthorized access [11]. To safeguard data privacy and security in 6G-enabled CEDs with Digital Twins, organizations must adopt robust security protocols, encompassing data encryption, authentication, and access control [5]. Moreover, instituting periodic security audits and risk assessments is crucial for identifying potential vulnerabilities and threats. Employing secure communication channels and data anonymization techniques further diminishes the risks associated with data privacy and security in this context. Through these measures, organizations can ensure the security and compliance of their metaverse systems with data privacy regulations, concurrently yielding valuable insights for enhanced decision-making and business operations.

To mitigate security risks associated with DT 6G-enabled CEDs in Metaverse, Intrusion Detection Systems (IDSs) [12], [13], [14], [15], [16] are employed to continuously analyze the access behavior of CEDs and identify potential attacks [17]. Anomaly-based Intrusion Detection Systems (IDS) categorize access behaviors into normal and anomalous based on the extent of deviation between observed behavior and the model [18], [19], [20], [21]. The adoption of Machine Learning (ML) and Deep Learning (DL) techniques has surged for creating IDS solutions capable of identifying malicious

Manuscript received 22 June 2023; revised 28 August 2023; accepted 1 October 2023. Date of publication 3 October 2023; date of current version 26 April 2024. This work was supported by the Researchers Supporting Project number (RSP2023r32), King Saud University, Riyadh, Saudi Arabia. (Corresponding author: M. Shamim Hossain.)

Suli He and Chengwen Du are with the Department of Electronics, Software Engineering Institute of Guangzhou, Guangzhou 510990, China (e-mail: ml15626131598_1@163.com; dxfaixp@163.com).

M. Shamim Hossain is with the Department of Software Engineering, College of Computer and Information Sciences, King Saud University, Riyadh 12372, Saudi Arabia (e-mail: mshossain@ksu.edu.sa).

Digital Object Identifier 10.1109/TCE.2023.3321846

threats [22]. However, aggregating data management across Consumer Electronics Devices (CEDs) could engender data privacy concerns. CED owners often resist sharing local data to preserve privacy, resulting in a data island dilemma that complicates the development of a robust intrusion detection model [23].

Therefore, the learning-based IDS model needs to be further developed to address the above two challenges. Federated Learning (FL) [24] is an efficient distributed learning approach that enables CEDs participating in training to jointly build a global intrusion detection model without disclosing local private data. FL enables devices to perform training on local data and send only local model parameters to the server for model aggregation [25], [26]. This approach establishes a parallel framework in the CED ecosystem for cooperative global intrusion detection model training while preserving data privacy [27].

Despite the privacy-preserving nature of FL in addressing the local data privacy concern, a significant challenge persists in the form of severe class imbalance during local training, as highlighted in the work by Wang et al. in [28]. Furthermore, within the heterogeneous FL-based metaverse environment, the Non-independent and Non-identically Distributed (non-IID) nature of training data, arising from variations in the proportions of local data among CEDs, has a substantial impact on the effectiveness of intrusion detection models, as noted by Zhao et al. in [29]. Consequently, applying existing FL approaches directly to intrusion detection in CEDs fails to produce intrusion detection models of high quality. Moreover, the training data held by each CED device exhibits a significant imbalance between normal and abnormal access behavior. Typically, normal behavior data dominates the training dataset, while abnormal behavior data is relatively scarce, resulting in a class imbalance problem [30]. Consequently, the bias introduced by this class imbalance during local training imposes limitations on the performance of IDSs.

Presently, the challenge of class imbalance is mitigated through the adoption of Imbalanced Learning (IL) algorithms, exemplified by He et al.'s ADASYN algorithm [31]. These methods seek to rectify bias through data resampling techniques, explored by Wang et al. in [32], or reweighting strategies, demonstrated by Duggal et al. [33]. However, these techniques are susceptible to instability and exhibit high computational demands in practical applications, as underscored by Japkowicz in [34]. Recently, meta-learning techniques have emerged as a promising avenue to surmount these limitations, as evidenced by Moniz and Cerqueira in [35]. Yet, these approaches' generalization capacity is limited due to their reliance on a model-specific optimization process. While certain studies have refined FedAvg [24] through modifications to the loss function and utilization of control variables to rectify client-side bias, these measures offer only partial remedies for non-IID data issues. Regrettably, these methods fall short in the realm of intrusion detection models within real-world scenarios, encompassing both local data class imbalance and non-i.i.d data dispersion across clients. Consequently, a comprehensive federated intrusion detection framework remains indispensable to navigate the complexities of heterogeneous metaverse environments.

In this paper, we propose a novel anomaly detection framework, rooted in federated meta-learning, tailored for intrusion detection in Digital Twin (DT) 6G-enabled Consumer Electronics Devices (CEDs) within metaverse environments. Our approach aims to tackle the challenges stemming from non-i.i.d. data distribution and class imbalance encountered in Intrusion Detection System (IDS) tasks. Specifically, our framework, called *FML-DT*, utilizes meta-learning and DT strategies to develop a sampling strategy for selecting the most effective samples to train intrusion detection models. The *FML-DT* approach enables clients to benefit from the meta-sampler provided by the server while ensuring data privacy by training the meta-sampler separately from local model training. This approach effectively mitigates the class imbalance problem of local data. Additionally, we propose a dynamic clustering and ranking algorithm for the model upload stage of FL. The algorithm clusters client-side models based on client-supplied information vectors, which consist of a set of local non-private information. It then evaluates the performance of each model and ranks them accordingly. This algorithm addresses the issue of global model performance degradation caused by the non-i.i.d nature of the data by dynamically removing local models with training state biases. Overall, our proposed approach provides a novel solution for intrusion detection in DT 6G-enabled CEDs, which can significantly enhance the performance and effectiveness of IDS systems in real-world scenarios.

The main contributions of this paper include:

- We propose a federated meta learning algorithm to address imbalance data in heterogeneous metaverse environments, which effectively mitigates the negative impact of imbalance data on intrusion detection performance.
- We leverage meta-learning in FL to construct a meta-sampler that learns the sampling strategy. The meta-sampler is optimized through reinforcement learning during the meta-training stage to continuously sample the balanced dataset on the client side. This approach significantly mitigates the impact of the class imbalance problem in the training data.
- We propose a federated clustering algorithm that addresses the non-i.i.d data problem by dynamically clustering and ranking client-side models using information vectors provided by clients. The proposed algorithm reduces the negative impact of extreme non-i.i.d data among clients on the FL training process, resulting in improved model performance.
- We conduct extensive evaluations on the intrusion detection datasets, and the experimental results demonstrate that our proposed framework is highly effective and produces intrusion detection models with superior performance.

II. RELATED WORKS

A. Intrusion Detection for DT 6G-Enabled Metaverse

With the increasing use of Digital Twin 6G-enabled CEDs in metaverse, the need for effective intrusion detection systems is paramount to ensure data privacy and security [18]. An IDS is designed to monitor network traffic and identify potential

security breaches or unauthorized access attempts [8]. Within the context of Digital Twin 6G-enabled CEDs, IDSs play a pivotal role in detecting anomalies within data flows and pinpointing system vulnerabilities. IDS employs various methodologies including signature-based, anomaly-based, or hybrid techniques for threat identification. Furthermore, the integration of machine learning and artificial intelligence algorithms enhances the detection prowess of IDS [10], [36].

Nevertheless, with the proliferation of CEDs and the exponential growth of Digital Twins, IDS face the challenge of managing and processing vast volumes of data in real-time to enable prompt alerting. Hence, it becomes essential to integrate IDS with cloud-based analytics and big data technologies, as highlighted in the work by Sasikumar et al. in [9], to ensure its efficacy. Through the implementation of a robust IDS, organizations can enhance their capability to identify and respond to potential security breaches or unauthorized access attempts, thereby fortifying the security and dependability of their Digital Twin 6G-enabled Metaverse systems. Among the most popular approaches to CEDs intrusion detection, machine learning-based methods have demonstrated great potential and promise [18]. For example, in [36], the authors introduced a new regularizer in traditional machine learning to address the challenges posed by high-dimensional, sparse, and lack of abnormal data in network data. In another study, the authors of [37] proposed a smart intrusion detection algorithm for IoT based on big data mining, using a combination of generative adversarial network (GAN) and convolutional neural network (CNN). However, these typical deep neural network methods require the aggregation of data from different devices for centralized training, which can raise serious privacy concerns.

To address these challenges, scholars have introduced a pioneering distributed model training framework known as Federated Learning (FL) [24]. In recent investigations, collaborative intrusion detection approaches grounded in FL have emerged, as exemplified by the blockchain-utilizing strategy proposed in [38] for secure storage of training models. Addressing intricate industrial systems, the DeepFed technique was advanced in [39] to discern cyber threats. Furthermore, [40] and [41] presented FL-based intrusion detection algorithms proactively identifying intrusions in CED networks. Despite their performance-centric focus, these methods have yet to adequately confront the prevalent issues of data class imbalance and non-i.i.d. disparities among clients encountered in authentic intrusion detection scenarios. Consequently, a more robust remedy is requisite to tackle these challenges effectively.

B. Imbalanced Learning

The challenges stemming from data class imbalance and non-i.i.d. data substantially impact the performance of FL-trained neural networks [28]. To combat data class imbalance among clients, a centralized monitoring system is proposed in [42], aimed at detecting class imbalance within training data. Monitors employ client-contributed partial data to estimate class distribution. Similarly, [32] presents a strategy to enhance the global model's accuracy by sharing training data

TABLE I
SUMMARY OF MAIN SYMBOLS

C_{update}	The cost of the meta-sampler to perform a single gradient update step
d	The dimensions of the dataset
$\mathcal{D}_k$	The local training dataset of k -th client
$\mathcal{D}^T$ and $\mathcal{D}^V$	The training and validation set
$\mathcal{D}_s$	The server-side training dataset for meta-sampler
E	The distribution of training or validation errors
$\hat{E}$	The histogram distribution of training or validation errors
n_k	The number of training samples of k -th client
N	The total number of samples in the training data set
ω	The global model
ω^k	The local model on client k
ℓ	The loss function
$\mathcal{S}$	The local sampler

among clients, thereby reducing the weight variance in FedAvg arising from non-i.i.d. data. Another avenue, detailed in [43], augments FedAvg's performance on non-i.i.d. data through a reinforcement learning-based client selection mechanism. The FL server, equipped with an agent, orchestrates client computations in each round, employing a reward function to evaluate the global model's validation data accuracy. Notwithstanding, these solutions entail client data sharing, thereby raising data privacy concerns [15], [19], [20]. Moreover, the extant research insufficiently addresses the effective resolution of the non-i.i.d. and class imbalance issues among client data in FL. Hence, our objective is to devise an FL-grounded intrusion detection system tailored for heterogeneous IoT environments, capably tackling these challenges within real-world scenarios.

III. PRELIMINARIES

A. Federated Learning

Federated Learning (FL) is an innovative collaborative machine learning paradigm that facilitates the collective training of models by decentralized participants, eliminating the necessity to share their sensitive data [44]. Within a typical supervised FL framework, a server denoted as $\mathcal{S}$ interacts with K clients. Each client holds a comprehensive labeled dataset $\mathcal{D}$, while the server lacks direct access to any individual client's data. The main objective of FL is to aggregate the locally trained models from all clients while maintaining data privacy, thereby obtaining a global model that exhibits comparable performance to centralized learning. One of the widely employed strategies in FL is FedAvg [24]. The FedAvg approach involves the distribution of the global model to each client, and the clients subsequently train their respective models locally using their available datasets. Following local model training, each client transmits its locally trained model back to the server. The server then aggregates the model parameters by averaging them and subsequently sends the updated global model back to the clients. This iterative cycle persists until the global model achieves convergence. By harnessing computational resources across multiple clients and amalgamating their models, FL achieves elevated model accuracy, concurrently upholding the confidentiality of individual client data. The description of the symbols used in the manuscript is summarized in Table I.

In practice, clients are permitted to communicate local model updates to the server, eliminating the necessity of data sharing while achieving the objective of collectively training

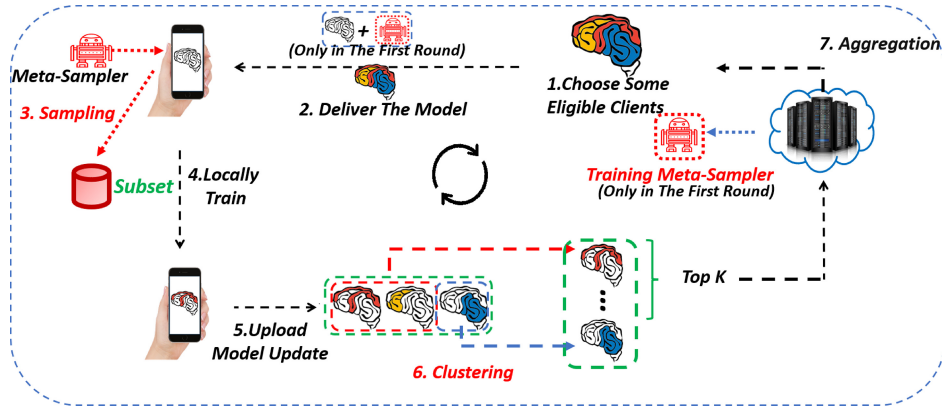


Fig. 1. Overview of the proposed system.

aggregated models. Consequently, the aim of FL centers around the minimization of the ensuing objective function:

$$f(\omega) = \sum_{k=1}^K \frac{n_k}{n} F_k(\omega), F_k(\omega) = \frac{1}{n_k} \sum_{i \in D_k} \ell_i(\omega), \quad (1)$$

Here, n represents the total sample count. To effect the minimization of the aforementioned objective function, a collaborative T -round training protocol is executed jointly by the server and clients. In this process, each client employs a local optimizer (such as SGD) on their local dataset D_k to refine the received global model ω_t , subsequently conveying the model updates to the server. Aggregating the updates furnished by clients, the server employs AGR algorithms to derive an optimized global model denoted as ω^* . This sequence of steps is reiterated over multiple rounds (e.g., T rounds) until the global model attains convergence.

B. Class Imbalance in FL

The issue of class imbalance constitutes a pervasive challenge across diverse real-world contexts, frequently resulting in marked deterioration of ML model performance. In the realm of FL, which operates within a decentralized framework, class imbalance introduces further complexities. These imbalances can manifest within individual client datasets or extend globally. Nonetheless, FL introduces distinct challenges attributed to the confidential nature of client data distributions, often safeguarded to uphold user privacy. Consequently, the task of rectifying class imbalance within the FL paradigm is notably intricate, diverging from conventional ML approaches.

In a recent investigation by [28], it is substantiated through mathematical analysis that class imbalance in training data can induce a reduction in the classification accuracy of FL models. Under the assumption $\hat{p}_{data} = p_{test}$, where p_{test} signifies the distribution of a balanced test set, and in the context of the loss function ℓ and the training data distribution $\hat{p}_{data}$, the optimal weight update for deep learning is expressed as follows:

$$w_{t+1}^* = w_t^* - \eta \nabla_{w_t^*} \sum_{i=1}^n \ell(f(x^{(i)}; w_t^*), y^{(i)}), \quad (2)$$

where $(x^{(i)}, y^{(i)}) \sim p_{test}$ and weight w^* denotes the weight vector that attains the highest accuracy on the test set and

serves as the optimal weight for FL. Consequently, the optimal objective for FL can be formulated as follows:

$$\min_{\omega} \mathbb{E}_{(x,y) \sim \hat{p}_{data}^{(k)}} \sum_{k=1}^K \ell[f(x; w^{(k)}), y], \quad (3)$$

In scenarios involving class imbalance, the distribution $\hat{p}_{data}^{(k)}$ of the training data for any client k can exhibit an imbalance, where $\hat{p}_{data}^{(k)}$ represents the data distribution of client k . In such situations, the weight update for the FL server can be expressed as follows:

$$\begin{aligned} \omega^{t+1} &= \sum_{k=1}^K \frac{n_k}{n} \omega_k^{t+1}, \\ &= \sum_{k=1}^K \frac{n_k}{n} \omega_k^t - \frac{\eta}{n} \nabla_{\omega_k^t} \sum_{i=1}^{n_k} \ell(f(x; \omega_k^t), y). \end{aligned} \quad (4)$$

It can be seen from the above formula that if the data is unbalanced, it is difficult for us to obtain the optimal solution, so the performance of the model may decline.

Moreover, the class imbalance issue in FL can be categorized into different types [45]: (i) Local imbalance, where a client's local dataset exhibits class imbalance if the ratio of the number of samples in the majority class to the number of samples in the minority class equals $IR = \max_p \{N_i^p\} / \min_p \{N_i^p\}$. (ii) Global imbalance, where an ensemble dataset can be formed by merging all client datasets, which has a ratio between the total number of samples in the majority class and the total number of samples in the minority class. This ratio is denoted as $IR = \max_p \{\sum_i N_i^p\} / \min_p \{\sum_i N_i^p\}$. (iii) Mismatch imbalance, where the imbalance of local data possessed by each client is distinct from the global data imbalance, and N_i^p refers to the number of data samples in class p on client i . This study focuses on the first two types of problems.

Discussion: The distributed nature of FL poses several challenges to model training, such as communication costs, system heterogeneity, and statistical heterogeneity. Statistical heterogeneity in FL arises due to the non-iid nature of training data, which is addressed to some extent by the techniques presented in Section II for intrusion detection data in FL. However, these techniques assume a balanced distribution of data classes,

which is not always the case in practical scenarios. While class imbalance in FL is considered a variant of the non-iid problem, this view is not entirely accurate. Firstly, the class imbalance is not limited to non-iid data scenarios. Secondly, in non-iid scenarios, the class imbalance is often caused by significant differences in the distribution ratio of each client class, leading to greater deviations in the FL model's training direction. Aggregating global models using biased local models results in biased models. Furthermore, as the non-iid degree of data increases, the trained model's performance significantly deteriorates. Experimental results show that traditional non-iid data solutions are ineffective in addressing class imbalance problems.

C. Clustering Algorithms in FL

Clustering, a data grouping technique, partitions data based on specific criteria into clusters or classes, aiming to enhance similarity among data points within the same cluster while minimizing dissimilarity among those in distinct clusters [46]. Clustering's core aim is to cohesively group data points of the same category while segregating those of disparate categories. In this context, we propose the application of clustering within Federated Learning (FL) to address the issue of statistical heterogeneity, which can contribute to the degradation of global model performance. We propose grouping clients based on similarities in their data distributions, whereby akin data distribution clients are allocated to the same cluster. This approach diminishes weight discrepancies within clusters at the edge device level. In the FL setting, we consider K clients and their respective local datasets denoted as $D_1, D_2, \dots, D_K$. We partition the K clients and datasets into k (where $k (k \leq K)$) distinct, non-overlapping clusters designated as $c_1, c_2, \dots, c_k$, wherein clients within the same cluster share akin data distributions. This can be achieved through a straightforward clustering process.

$$\min_{\omega^1, \omega^2, \dots, \omega^K} \sum_{k=1}^K \frac{1}{\sum_{l \in c_k} n_l} \sum_{i \in c_k} n_i \sum_{j=1}^{n_i} \frac{1}{n_i} \ell(\omega^k; D_K). \quad (5)$$

D. Problem Definition

Digital Twin 6G-enabled Metaverse systems have introduced a groundbreaking paradigm across diverse domains, encompassing smart cities, autonomous vehicles, and industry 4.0 [47]. Notwithstanding their transformative potential, the escalating complexity and heterogeneity of these systems usher forth substantial cybersecurity challenges, with intrusion attacks at the forefront. IDSs have surfaced as potent remedies for detecting and averting such attacks. Nevertheless, in the realm of Metaverse environments where data is dispersed among numerous CEDs and privacy considerations restrict data exchange, traditional IDS methodologies dependent on extensive data may prove impractical.

In this work, our objective is to develop a meta-learning approach for intrusion detection in FL-based Digital Twin 6G-enabled CEDs. Our proposed approach harnesses knowledge acquired from similar IDS tasks to enhance the performance

of the target IDS task. Specifically, we adopt a meta-learning framework where the model is initially trained on a set of source IDS tasks and subsequently adapted to a target IDS task with limited labeled data available. This approach enables the leveraging of prior knowledge to improve the effectiveness of intrusion detection in FL-based Digital Twin 6G-enabled CEDs, where data availability and privacy constraints are critical considerations.

To this end, we formulate the following research questions:

- How can we design a meta-learning approach for intrusion detection in FL-based Digital Twin 6G-enabled CEDs?
- How can we transfer knowledge from source IDS tasks to improve the performance of the target IDS task with limited data?
- How does the proposed approach compare to state-of-the-art IDS methods in terms of detection accuracy, false alarm rate, and training efficiency?

This study endeavors to surmount these challenges through the introduction of an innovative meta-learning methodology for intrusion detection within FL-based Digital Twin 6G-enabled CEDs. Our approach capitalizes on shared attributes among diverse IDS tasks, thereby augmenting the performance of the target IDS task. We postulate that our method can proficiently identify intrusion attacks even in scenarios characterized by scarce labeled data. Furthermore, our objectives encompass curtailing communication overhead within FL while upholding paramount data privacy considerations. By harnessing meta-learning's potential, our approach is poised to facilitate the efficient detection of intrusion attacks in FL-based Digital Twin 6G-enabled CEDs. We anticipate its efficacy in mitigating the challenges posed by limited labeled data, enriching IDS performance, and abating potential privacy apprehensions. This research contributes to the evolution of intrusion detection techniques within the realm of FL-based Digital Twin 6G-enabled CEDs, ultimately heightening the security and dependability of these systems.

IV. METHODOLOGY

A. Meta-Training and Meta-Sampling

In this section, we will present a comprehensive explanation of the meta-training stage of the server and the meta-sampling stage of the client, utilizing Algorithm 1 and Algorithm 2, respectively.

(a) Meta-state. To ensure data privacy within FL, it becomes imperative to render the training phase of samplers independent from the FL training phase, while circumventing the usage of client-specific local data. Consequently, the identification of a task-agnostic state that imparts guidance for meta-sampler training without reliance on the client's local data is paramount. To surmount this challenge, we present a novel approach, inspired by prior research [48], wherein we employ the histogram distribution of training and validation errors as the guiding meta-state for meta-sampler training. This meta-state encompasses histograms of both training and validation errors. Concretely, the vector $\hat{E}_D$ is derived utilizing

Algorithm 1 Meta-Sampling Algorithm**Input:** Majority set $\mathcal{P}_\tau$ and minority set $\mathcal{N}_\tau$ from D_τ ;**Output:** Balanced subset $\mathcal{D}'_\tau = \mathcal{N}_\tau \cup \mathcal{P}_\tau$;

- 1: Assign each (x_i, y_i) in $\mathcal{N}_\tau$ with weight:

$$w_i = \frac{g_{\mu, \sigma}(|F(x_i) - y_i|)}{\sum_{(x_j, y_j) \in \mathcal{N}_\tau} g_{\mu, \sigma}(|F(x_j) - y_j|)}$$
- 2: Sample majority subset $\mathcal{N}'_\tau$ from $\mathcal{N}_\tau$ sampling weights w , where $|\mathcal{N}'_\tau| = |\mathcal{P}_\tau|$
- 3: **return** Balanced subset $\mathcal{D}'_\tau = \mathcal{N}'_\tau \cup \mathcal{P}_\tau$

the following formula:

$$\hat{E}_D^i = \frac{\left| \left\{ (x, y) \mid \frac{i-1}{b} \leq \text{abs}(F_i(x) - y) \leq \frac{i}{b}, (x, y) \in D \right\} \right|}{|D|}, \quad (6)$$

where $1 \leq i \leq b$ and (x, y) denotes a data point, and the classification error e is defined as the absolute difference between the predicted probability that x is positive and the label y , i.e., $|F_i(x) - y|$. The i -th component of E_D is denoted as $\hat{E}_D^i$, and the number of bins in the histogram is represented as b .

(b) Meta-sampling. To streamline the sampling process and accommodate the computational capacities of CEDs, we refrain from employing an extensive output layer within the meta-sampler to ascertain the sampling probability of data samples. Instead, we streamline the procedure through the adoption of a Gaussian function:

$$g_{\mu, \sigma}(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2}, \quad (7)$$

where e represents Euler's number, and the meta-sampler calculates $\mu \in [0, 1]$ while σ is a hyperparameter.

According to Algorithm 1, the meta-sampling stage can be described as follows: During the meta-sampling phase, the meta-sampler $\mathcal{S}$ first produces a scalar μ by processing the input meta-state, which is represented as $\mu \sim \mathcal{S}(\mu|s)$. Then, the sampling weight of each data sample is determined by applying the Gaussian function discussed earlier to its classification error, as indicated in line 2 of the algorithm. Lastly, the data samples from the majority set $\mathcal{P}_\tau$ are sampled using these weights to obtain a balanced dataset.

(c) Meta-training. To address the class imbalance problem in FL, we propose to train a meta-sampler $\mathcal{S}$ to obtain a balanced dataset by iteratively sampling data. The meta-state s of the client serves as input to $\mathcal{S}$, which outputs a scalar μ to guide the Gaussian function for calculating the sampling weight of each data sample. It should be noted that training the non-differentiable meta-sampler s is an optimization problem, which we solve using reinforcement learning (RL). Algorithm 2 presents the pseudocode for this approach. At each episode, the classifier F is trained on a randomly balanced subset. Then, at each step, the current training state s_t is used to obtain the action μ from the meta-sampler. Using this μ , the meta-sampling stage obtains a balanced subset $\mathcal{D}'_{t+1}$, which is used to update the classifier F to F_{t+1} , and obtain a new state s_{t+1} . We define the reward in RL as the performance difference between F_t and F_{t+1} on the validation set, using an evaluation function A . The optimization goal of the meta-sampler is

Algorithm 2 Federated Learning-Based Meta-Sampling Algorithm**Input:** The server-side model F_s , the server-side dataset $\mathcal{D}_s$, the built meta-sampler $\mathcal{Q}$, and the parameters in [49] for RL.**Output:** Meta-sampler $\mathcal{Q}$ **At the server:**

- 1: **for** $step = 1, \dots, N$ **do**
- 2: Initializes ω_s
- 3: Cuts $\mathcal{D}_s$ into the training set $\mathcal{D}_s^T$ and validation set $\mathcal{D}_s^V$
- 4: **for** Meta training $step = 1, \dots, n$ **do**
- 5: Computes $\hat{E}_{\mathcal{D}_s^T}$
- 6: Computes $\hat{E}_{\mathcal{D}_s^V}$
- 7: Builds meta-state $s_n = [\hat{E}_{\mathcal{D}_s^T}; \hat{E}_{\mathcal{D}_s^V}]$
- 8: Takes action from $\mathcal{Q}(\mu_t|s_n)$, i.e., $\mu_n \sim \mathcal{Q}(\mu_t|s_n)$
- 9: Uses Algorithm 1 to obtain $\mathcal{D}'_n$ $\backslash \backslash$ refer to

Algorithm 1

- 10: $\omega_s^{n+1} = \omega_s^n(\mathcal{D}'_n)$
- 11: Observes reward $r_n = \mathcal{A}(\omega_s^{n+1}, \mathcal{D}_s^V) - \mathcal{A}(\omega_s^n, \mathcal{D}_s^V)$
- 12: Computes s_{n+1} with ω_s^{n+1}
- 13: Stores transition $\mathcal{M} = \mathcal{M} \cup (s_n, \mu_n, r_n, s_{n+1})$
- 14: **end for**
- 15: **for** each gradient step **do**
- 16: Update $\psi, \bar{\psi}, \theta$, and $\omega_{\mathcal{Q}}$ according to [49]
- 17: **end for**
- 18: **end for**

to sample high-quality data for training a high-performance model.

For optimization of the meta-sampling process $\mathcal{S}$ using Reinforcement Learning (RL) techniques, we use the Soft Actor-Critic (SAC) algorithm [50]. SAC stands as an off-policy deep RL algorithm that functions within the maximum entropy RL framework. It encompasses multiple advantages over alternative RL approaches, encompassing heightened exploration capacities, augmented resilience, and adeptness in adapting amidst interference. Furthermore, SAC accelerates training due to its adherence to the maximum entropy principle, which promotes more balanced exploration. The SAC algorithm consists of several key components, such as the state value function $V_\psi(s_t)$, the target network $V_{\bar{\psi}}(s_t)$, the soft Q function $Q_\theta(s_t, a_t)$, and the tractable policy $\mathcal{S}_\phi(a_t|s_t)$. The update of the corresponding parameters can be found in line 12 of Algorithm 2.

Complexity of Federated Meta Sampling: Here, we assume that the cost of the meta-sampler to perform a single gradient update step is C_{update} (depending on the gradient computation formula). In our implementation, following the SAC optimization strategy, we need to perform n_{random} actions before updating the meta-sampler and then need n_{update} steps to collect online transitions and perform gradient updates for $\xi(\cdot)$. Therefore, the overall cost of meta-training can be expressed as $O((n_{random} + n_{update})NBC) + O(n_{update}C_{update})$. For training, we assume that there are c clients joining, then the overall time complexity of the proposed attack is $\mathcal{O}(cN \log c(\log \kappa + dBC)) + O(c \log c((n_{random} + n_{update})NBC + n_{update}C_{update}))$.

B. Federated Clustering Algorithm

In this section, we explore the use of clustering algorithms in FL. However, current methods typically require clustering based on the data distribution of clients, which compromises the privacy of the clients and makes them more vulnerable to attacks. To address this issue, we propose a novel federated clustering method suitable for CEDs that clusters clients based on a small amount of non-private data.

(a) Information Vector. To address privacy concerns in FL, a vector-based data representation is proposed to replace the traditional data distribution for clustering. This new representation enables the transmission of lightweight information without compromising client privacy. The aim is to identify a set of data that accurately represents the current client state in a lightweight manner. The proposed approach utilizes three pieces of data, namely the client's imbalance ratio (IR), training accuracy, and validation accuracy, to construct a vector. Under non-i.i.d data partitioning, each client has a unique imbalance ratio that characterizes their data's class distribution. Training and validation accuracy are selected to capture each client's training state information, based on the theory discussed in meta-state. It is believed that the constructed information vector accurately represents each client's unique class distribution and training state information.

(b) Clustering and Ranking Algorithm. Subsequent to the clustering process using the information vector discussed above, the uploaded models from the clients in the k -th cluster are aggregated to obtain the aggregation model ω_k for that cluster. This leads to the formation of an aggregated model for each category. To filter out the models with subpar training performance, a ranking algorithm has been proposed. The ranking algorithm is formulated as follows:

$$\forall \omega_{i_n} \leftarrow \max \sum_{j=1}^n P(\omega_{i_j}), \quad 8$$

where $i_j \in (1, k)$, $i_1 \neq i_2 \dots \neq i_n$. According to this algorithm, we rank the obtained aggregation models and finally obtain a set of aggregation models with the best performance. Here, i_j represents the number of aggregation models, and each model is selected only once during the ranking process ($i_1 \neq i_2, \dots \neq i_n$). $P(\cdot)$ is the evaluation function, and n is used as a hyperparameter to control the number of models after ranking.

Algorithm 3 outlines the workings of the dynamic clustering ranking algorithm as follows: in each epoch, the server collects the information vector from each client and employs it for clustering to determine the cluster assignment of each client. It is noteworthy that the clients provide the information vector at every upload stage, resulting in frequent clustering. Since the training status of the clients changes with each iteration, their clustering also changes correspondingly, making the entire clustering ranking process a continuous dynamic adjustment. The frequency of uploading the information vector can be customized based on the requirements of the FL system. Clients belonging to the same cluster submit updated models, which

Algorithm 3 Federated Learning-Based Clustering Algorithm

Input: The local model ω , the given number of clusters k , the ranking threshold r , and information vector $\mathcal{I}$;

Output: Clustered aggregation model ω_c ;

```

1: for each client  $C_i$  do
2:    $\mathcal{I}_i \sim C_i$ 
3:   Update  $\mathcal{I}$  with  $\mathcal{I}_i$ 
4: end for
5:  $c = \text{Clustering Algorithm}(\mathcal{I}, k)$ 
6: for  $c_i$  in  $c$  do
7:    $\omega_{c_i} \sim \text{Aggregation}(\omega_i \in c_i)$ 
8: end for
9:  $\omega_c = [\omega_{c_1}, \omega_{c_2}, \omega_{c_3} \dots \omega_{c_n}]$ 
10:  $\omega_c = \text{Ranking}(\omega_c, r)$ 
11: return  $\omega_c$ 

```

are aggregated in step 7. Finally, employing the above equations, the top- k aggregation models with the best performance are selected.

V. EXPERIMENT

A. Experiment Setup

1) *Datasets and Splits:* The **UNSW-NB-15** and **NSL-KDD** datasets are two well-known benchmark data sets commonly used to evaluate intrusion detection systems (IDS) based on network traffic data. The UNSW-NB-15 dataset was created by the University of New South Wales (UNSW) in Australia and contains 2.5 million network traffic instances captured in a simulated environment. The dataset includes various attack categories, such as denial-of-service (DoS), user-to-root (U2R), remote-to-local (R2L), and probing attacks. It is considered one of the most comprehensive datasets for evaluating network-based IDS, particularly for deep learning models. The NSL-KDD dataset is a modification of the original KDD Cup 99 dataset, which was created for the 1999 KDD Cup competition. The NSL-KDD dataset was created to address some of the limitations of the KDD Cup 99 dataset, such as the lack of diversity in the attack types and the presence of redundant and irrelevant features. The NSL-KDD dataset contains approximately 1.8 million network traffic instances, with four attack categories and several normal traffic instances. It is commonly used for evaluating both traditional and deep learning-based IDS approaches. To simulate number-balanced and class-unbalanced non-IID settings, we partition the training set into multiple subsets, each corresponding to a different client, and then assign each client a different number of samples from each class.

2) *Training Details:* In our experimental setup, we consider a scenario with a total of $K = 100$ clients. In each communication round, the server randomly selects 10 clients to participate in the training process. We conduct a total of $T = 150$ or $T = 50$ iterations, depending on the specific experiment. For the hyperparameters, we set the *Beta* distribution parameter δ to 0.75, the threshold τ to 0.80, the learning rate η to 0.01, and the local epoch to 5. These values determine the weights assigned to the client and server models during

TABLE II
RESULTS OF IDS ON THE UNSW-NB 15 DATASET

Dataset	UNSW-NB-15				
Method	FedAvg	FedProx	Over-sampling	Under-sampling	Ours
IR=20%	0.44	0.44	0.75	0.74	0.82
IR=40%	0.40	0.44	0.75	0.76	0.84
IR=60%	0.44	0.44	0.75	0.75	0.83
IR=80%	0.44	0.44	0.75	0.76	0.82

TABLE III
RESULTS OF IDS ON THE NSL-KDD DATASET

Dataset	UNSW-NB-15				
Method	FedAvg	FedProx	Over-sampling	Under-sampling	Ours
IR=20%	0.68	0.68	0.71	0.75	0.83
IR=40%	0.67	0.48	0.73	0.76	0.82
IR=60%	0.67	0.49	0.72	0.74	0.83
IR=80%	0.67	0.43	0.73	0.76	0.83

the aggregation process. The weight of the client model is $\alpha = 0.4$, the weight of the server model is $\beta = 0.4$, and the weight of the global model from the previous round is $\gamma = 0.2$. These weights control the influence of each model on the updated global model. To optimize the ResNet-9 ML model, we employ the Adam optimization algorithm with an initial learning rate of $\eta = 1e - 3$. The client training process consists of a single epoch ($\mathcal{E}_u = 1$) with a batch size of $\mathcal{B}_u = 128$. Similarly, the server training process involves one epoch ($\mathcal{E}_s = 1$) with a batch size of $\mathcal{B}_s = 128$.

B. Experimental Results

1) *IDS Tasks*: Table II and Table III demonstrate the IDS performance of the proposed framework on the UNSW-NB 15 and NSL-KDD datasets. From the experimental results, it can be seen that the proposed framework outperforms all baselines and is more stable than them. This is mainly because the proposed framework uses a meta-learning strategy to adaptively sample efficient samples to improve performance. We can also know that the FedProx algorithm used to solve the non-IID problem is not suitable for the class imbalance scenario.

2) *Ablation Analysis*: The results in Table IV demonstrate the impact of using the dynamic clustering and ranking algorithm on the performance of the FML-DT framework. The comparison between using and not using the dynamic clustering ranking algorithm in two cases, and the F1-Score, AUC, Recall, and Precision are evaluated. It can be observed that using the algorithm leads to the higher detection performances under the non-IID conditions. When the algorithm is not used, the models provided by some clients with the non-IID data and class imbalance may cause the global model performance to degrade during the model aggregation. Moreover, even without using the dynamic clustering and ranking algorithms, FML-DT's detection performance is still close to the baseline method shown in Table II and Table III. By utilizing the dynamic clustering and ranking algorithm, it is observed improved detection performance. This is crucial for the CEDs intrusion detection, and it highlights the importance of the proposed algorithm in FML-DT.

TABLE IV
COMPARING THE PERFORMANCE OF FML-DT WITH AND WITHOUT THE PROPOSED FEDERATED CLUSTER RANKING ALGORITHM

Metric	Ours	w/o-dynamic cluster ranking
F1-Score	83.69	72.50
AUC	83.05	75.49
Recall	78.96	61.30
Precision	89.03	88.71

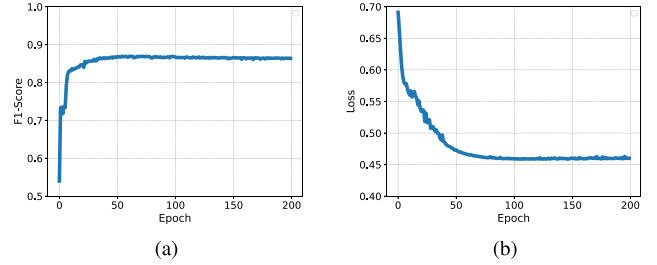


Fig. 2. Convergence performance of the proposed FML-DT.

3) *Convergence Analysis*: In the initial experiment, we first examined the convergence of FML-DT with the assumption that there are 100 idle IoT devices. The convergence performance of FML-DT is depicted in Fig. 2. It reveals that the F1 score and loss rapidly converge to near-optimal values within 500 training epochs. Furthermore, as the number of training iterations increases, it shows that there is a gradual improvement in the F1 score and a reduction in loss. It also indicates that the training process does not exhibit overfitting. Therefore, we can confidently state that our proposed FML-DT exhibits fine convergence performance.

VI. CONCLUSION

We proposed an innovative meta-learning approach for intrusion detection in the FL-based metaverse for digital twin 6G-enabled CEDs. The proposed approach addresses the challenge of class imbalance by training a meta-sampler to iteratively sample data and obtain a balanced dataset. The use of meta-learning also enables the training of the meta-sampler without relying on the client's local data, ensuring data privacy protection. The experiments conducted on the UNSW-NB-15 and NSL-KDD datasets demonstrated the effectiveness of the proposed approach in achieving high accuracy while maintaining data privacy.

REFERENCES

- [1] M. Choi, A. Azzaoui, S. K. Singh, M. M. Salim, S. R. Jeremiah, and J. H. Park, "The future of metaverse: Security issues, requirements, and solutions," *Human-Centric Comput. Inf. Sci.*, vol. 12, pp. 1–15, Dec. 2022.
- [2] H. Peng, P.-C. Chen, P.-H. Chen, Y.-S. Yang, C.-C. Hsia, and L.-C. Wang, "6G toward metaverse: Technologies, applications, and challenges," in *Proc. APWCS*, 2022, pp. 6–10.
- [3] A. Mukherjee, D. De, N. Dey, R. G. Crespo, and E. Herrera-Viedma, "DisastDrone: A disaster aware consumer Internet of Drone Things system in ultra-low latent 6G network," *IEEE Trans. Consum. Electron.*, vol. 69, no. 1, pp. 38–48, Feb. 2023.
- [4] M. Al-Hawawreh and M. S. Hossain, "Federated learning-assisted distributed intrusion detection using mesh satellite nets for autonomous vehicle protection" *IEEE Trans. Consum. Electron.*, early access, Sep. 25, 2023, doi: [10.1109/TCE.2023.3318727](https://doi.org/10.1109/TCE.2023.3318727).

- [5] Y. Liu, X. Yuan, Z. Xiong, J. Kang, X. Wang, and D. Niyato, "Federated learning for 6G communications: Challenges, methods, and future directions," *China Commun.*, vol. 17, no. 9, pp. 105–118, Sep. 2020.
- [6] L. U. Khan, W. Saad, D. Niyato, Z. Han, and C. S. Hong, "Digital-twin-enabled 6G: Vision, architectural trends, and future directions," *IEEE Commun. Mag.*, vol. 60, no. 1, pp. 74–80, Jan. 2022.
- [7] F. Lin, Y. Zhou, X. An, I. You, and K.-K. R. Choo, "Fair resource allocation in an intrusion-detection system for edge computing: Ensuring the security of Internet of Things devices," *IEEE Consum. Electron. Mag.*, vol. 7, no. 6, pp. 45–50, Nov. 2018.
- [8] R. Minerva, G. M. Lee, and N. Crespi, "Digital twin in the IoT context: A survey on technical features, scenarios, and architectural models," *Proc. IEEE*, vol. 108, no. 10, pp. 1785–1824, Oct. 2020.
- [9] A. Sasikumar et al., "Blockchain-based trust mechanism for digital twin empowered Industrial Internet of Things," *Future Gener. Comput. Syst.*, vol. 141, pp. 16–27, Apr. 2023.
- [10] K. M. Alam and A. El Saddik, "C2PS: A digital twin architecture reference model for the cloud-based cyber-physical systems," *IEEE Access*, vol. 5, pp. 2050–2062, 2017.
- [11] H. X. Nguyen, R. Trestian, D. To, and M. Tatipamula, "Digital twin for 5G and beyond," *IEEE Commun. Mag.*, vol. 59, no. 2, pp. 10–15, Feb. 2021.
- [12] E. Biermann, E. Cloete, and L. M. Venter, "A comparison of intrusion detection systems," *Comput. Security*, vol. 20, no. 8, pp. 676–683, 2001.
- [13] E. Kabir, J. Hu, H. Wang, and G. Zhuo, "A novel statistical technique for intrusion detection systems," *Future Gener. Comput. Syst.*, vol. 79, pp. 303–318, Feb. 2018.
- [14] V. Truong and L. B. Le, "MetaCIDS: A metaverse collaborative intrusion detection system based on blockchain and federated learning," 2023. [Online]. Available: <https://doi.org/10.36227/techrxiv.22816568.v2>
- [15] A. Makkar, T. W. Kim, A. K. Singh, J. Kang, and J. H. Park, "SecureIIoT environment: Federated learning empowered approach for securing IIoT from data breach," *IEEE Trans. Ind. Informat.*, vol. 18, no. 9, pp. 6406–6414, Sep. 2022.
- [16] E. Tcydenova, T. W. Kim, C. Lee, and J. H. Park, "Detection of adversarial attacks in AI-based intrusion detection systems using explainable AI," *Human-Centric Comput. Inform. Sci.*, vol. 11, pp. 1–14, Sep. 2021.
- [17] C. Alcaraz and J. Lopez, "Digital twin: A comprehensive survey of security threats," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 3, pp. 1475–1503, 3rd Quart., 2022.
- [18] P. Garcia-Teodoro, J. Diaz-Verdejo, G. Maciá-Fernández, and E. Vázquez, "Anomaly-based network intrusion detection: Techniques, systems and challenges," *Comput. Security*, vol. 28, nos. 1–2, pp. 18–28, 2009.
- [19] X. Wang, W. Liu, H. Lin, J. Hu, K. Kaur, and M. S. Hossain, "AI-empowered trajectory anomaly detection for intelligent transportation systems: A hierarchical federated learning approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 4631–4640, Apr. 2023.
- [20] V. Rey, P. M. S. Sánchez, A. H. Celdrán, and G. Bovet, "Federated learning for malware detection in IoT devices," *Comput. Netw.*, vol. 204, Feb. 2022, Art. no. 108693.
- [21] K. Yadav, B. B. Gupta, C.-H. Hsu, and K. T. Chui, "Unsupervised federated learning based IoT intrusion detection," in *Proc. GCCE*, 2021, pp. 298–301.
- [22] H. Liu and B. Lang, "Machine learning and deep learning methods for intrusion detection systems: A survey," *Appl. Sci.*, vol. 9, no. 20, p. 4396, 2019.
- [23] N. Shone, T. N. Ngoc, V. D. Phai, and Q. Shi, "A deep learning approach to network intrusion detection," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 2, no. 1, pp. 41–50, Feb. 2018.
- [24] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. AISTATS*, 2017, pp. 1273–1282.
- [25] S. Agrawal et al., "Federated learning for intrusion detection system: Concepts, challenges and future directions," *Comput. Commun.*, vol. 195, pp. 1273–1282, Nov. 2022.
- [26] Y. Zheng, S. Lai, Y. Liu, X. Yuan, X. Yi, and C. Wang, "Aggregation service for federated learning: An efficient, secure, and more resilient realization," *IEEE Trans. Depend. Secure Comput.*, vol. 20, no. 2, pp. 988–1001, Mar./Apr. 2022.
- [27] S. A. Rahman, H. Tout, C. Talhi, and A. Mourad, "Internet of Things intrusion detection: Centralized, on-device, or federated learning?" *IEEE Netw.*, vol. 34, no. 6, pp. 310–317, Nov./Dec. 2020.
- [28] L. Wang, S. Xu, X. Wang, and Q. Zhu, "Addressing class imbalance in federated learning," in *Proc. AAAI*, 2021, pp. 1–9.
- [29] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-IID data," 2018, *arXiv:1806.00582*.
- [30] M. Duan, D. Liu, X. Chen, R. Liu, Y. Tan, and L. Liang, "Self-balancing federated learning with global imbalanced data in mobile systems," *IEEE Trans. Parallel Distrib. Syst.*, vol. 32, no. 1, pp. 59–71, Jan. 2021.
- [31] H. He, Y. Bai, E. A. Garcia, and S. Li, "ADASYN: Adaptive synthetic sampling approach for imbalanced learning," in *Proc. IEEE Int. Joint Conf. Neural Netw.*, 2008, pp. 1322–1328.
- [32] S. Wang, L. L. Minku, and X. Yao, "Resampling-based ensemble methods for Online class imbalance learning," *IEEE Trans. Knowl. Data Eng.*, vol. 27, no. 5, pp. 1356–1368, May 2015.
- [33] R. Duggal, S. Freitas, S. Dhamnani, D. H. Chau, and J. Sun, "HAR: Hardness aware reweighting for imbalanced datasets," in *Proc. Big Data*, 2021, pp. 1–11.
- [34] N. Japkowicz, "Assessment metrics for imbalanced learning," *Imbalanced Learning: Foundations, Algorithms, and Applications*. New York, NY, USA: Wiley, 2013, pp. 187–206.
- [35] N. Moniz and V. Cerqueira, "Automated imbalanced classification via meta-learning," *Expert Syst. Appl.*, vol. 178, Sep. 2021, Art. no. 115011.
- [36] M. Almseidin, M. Alzubi, S. Kovacs, and M. Alkasasbeh, "Evaluation of machine learning algorithms for intrusion detection system," in *Proc. SISY*, 2017, pp. 277–282.
- [37] E. Seo, H. M. Song, and H. K. Kim, "GIDS: GAN based intrusion detection system for in-vehicle network," in *Proc. 16th Annu. Conf. Privacy, Security Trust (PST)*, 2018, pp. 1–6.
- [38] H. Liu et al., "Blockchain and federated learning for collaborative intrusion detection in vehicular edge computing," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 6073–6084, Jun. 2021.
- [39] B. Li, Y. Wu, J. Song, R. Lu, T. Li, and L. Zhao, "DeepFed: Federated deep learning for intrusion detection in industrial cyber-physical systems," *IEEE Trans. Ind. Informat.*, vol. 17, no. 8, pp. 5615–5624, Aug. 2021.
- [40] T. D. Nguyen, P. Rieger, M. Miettinen, and A.-R. Sadeghi, "Poisoning attacks on federated learning-based IoT intrusion detection system," in *Proc. DISS*, 2020, pp. 1–7.
- [41] M. Abdel-Basset, N. Moustafa, H. Hawash, I. Razzak, K. M. Sallam, and O. M. Elkomy, "Federated intrusion detection in blockchain-based smart transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 3, pp. 2523–2537, Mar. 2022.
- [42] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, Sep. 2009.
- [43] Z. Tang et al., "Data resampling for federated learning with non-IID labels," in *Proc. IJCAI*, 2021, pp. 1–7.
- [44] Y. Liu et al., "Deep anomaly detection for time-series data in industrial IoT: A communication-efficient on-device federated learning approach," *IEEE Internet Things J.*, vol. 8, no. 8, pp. 6348–6358, Apr. 2021.
- [45] Z. Shen, J. Cervino, H. Hassani, and A. Ribeiro, "An agnostic approach to federated learning with class imbalance," in *Proc. ICLR*, 2022, pp. 1–12.
- [46] A. Ghosh, J. Chung, D. Yin, and K. Ramchandran, "An efficient framework for clustered federated learning," in *Proc. NeurIPS*, 2020, pp. 19586–19597.
- [47] R. Gámez Díaz, Q. Yu, Y. Ding, F. Laamarti, and A. El Saddik, "Digital twin coaching for physical activities: A survey," *Sensors*, vol. 20, no. 20, p. 5936, 2020.
- [48] J. Shu et al., "Meta-weight-net: Learning an explicit mapping for sample weighting," in *Proc. NeurIPS*, 2019, pp. 1917–1928.
- [49] Y. Liu, J. J. Q. Yu, J. Kang, D. Niyato, and S. Zhang, "Privacy-preserving traffic flow prediction: A federated learning approach," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7751–7763, Apr. 2020.
- [50] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proc. ICML*, 2018, pp. 1856–1865.